

Paper Practice 2.4 • WORKED KEY**Solving Equations with the Variable on Each Side**

WORKED KEY (odd-numbered items) – Distribute first, collect the variable terms on one side, then solve. If the variables cancel, the equation is either never true (no solution) or always true (identity). Even-numbered items are yours to work.

Solve each equation. Check your solution.

1. $6v - 4 = -3v + 59$

Answer: $9v - 4 = 59 \rightarrow 9v = 63 \rightarrow v = 7$. Check: $42 - 4 = 38$ and $-21 + 59 = 38$ ✓

2. $-t + 7 = 3t + 15$

3. $8b + 5 = 12b - 27$

Answer: $5 + 27 = 12b - 8b \rightarrow 32 = 4b \rightarrow b = 8$. Check: $69 = 69$ ✓

4. $-4w + 3 = w - 27$

5. $\frac{2n+2}{8} = \frac{n}{3}$

Answer: cross-multiply: $3(2n + 2) = 8n \rightarrow 6n + 6 = 8n \rightarrow n = 3$. Check: $8 \div 8 = 1$ and $3 \div 3 = 1$ ✓

6. $\frac{4x+10}{2} = -3x$

7. $9(g - 4) = 2g - 1$

Answer: $9g - 36 = 2g - 1 \rightarrow 7g = 35 \rightarrow g = 5$. Check: $9(1) = 9$ and $10 - 1 = 9$ ✓

8. $-5(-a - 1) = 3a - 7$

9. $6 + 3(y + 4) = 2(y + 1) + 11$

Answer: $3y + 18 = 2y + 13 \rightarrow y = -5$. Check: $6 + 3(-1) = 3$ and $2(-4) + 11 = 3$ ✓

10. $8(q - 2) = 44 + 5(2q - 12)$

11. **PUMPKINS** Kelly and Gemma each pick a pumpkin from the pumpkin patch. Kelly's pumpkin weighs 5 pounds more than twice the weight of Gemma's pumpkin. Kelly's pumpkin also weighs 25 pounds less than 4 times the weight of Gemma's pumpkin. Write and solve an equation to find the weights of the pumpkins.

Answer: Let g = Gemma's weight. Both describe Kelly: $2g + 5 = 4g - 25 \rightarrow 30 = 2g \rightarrow g = 15$. **Gemma's pumpkin: 15 lb; Kelly's: $2(15) + 5 = 35$ lb.** Check: $4(15) - 25 = 35$ ✓

12. **SNOWFALL** In February 2015, South Bend, Indiana, saw one of its snowiest Februarys on record. The snowfall in February 2015 was 0.6 inch more than twice the snowfall in February 2020. The snowfall in February 2015 was also 8.35 inches fewer than 2.5 times the snowfall in February 2020. Write and solve an equation to find the amount of snowfall in February 2015 and in February 2020.

Example 5

13. **GEOMETRY** Write and solve an equation to find the value of x so that the figures have the same area.

(rectangle $x + 3$ by 2; rectangle $x - 1$ by 4)

Answer: set the areas equal: $2(x + 3) = 4(x - 1) \rightarrow 2x + 6 = 4x - 4 \rightarrow 10 = 2x \rightarrow x = 5$. Check: $2(8) = 16$ and $4(4) = 16$ ✓

14. **GEOMETRY** Write and solve an equation to find the value of x so that the figures have the same area.

(rectangle $x - 1$ by 2; rectangle $x - 4$ by 4)

Solve each equation, and state whether the equation has one solution, no solution, or is an identity.

15. $5p + 7 = -7 + 5p$

Answer: subtract $5p$ from both sides: $7 = -7$, which is never true $\rightarrow$ **no solution**

16. $4(x + 3) = 4x + 12$

17. $-9h + 11 = 3(3h + 6)$

Answer: $-9h + 11 = 9h + 18 \rightarrow -7 = 18h \rightarrow h = -\frac{7}{18}$; **one solution.** Check: both sides equal 14.5 ✓

18. $6b + 5 - 2b = 4(b + 5)$

19. $6(a + 2) = 3(1 - 2a)$

Answer: $6a + 12 = 3 - 6a \rightarrow 12a = -9 \rightarrow a =$
 $-\frac{3}{4}$; **one solution.** Check: $6(1.25) = 7.5$ and $3(1$
 $+ 1.5) = 7.5$ ✓

20. $9y + 8 = 4(y + 2) + 5y$